MUHAMMAD ZEIN AULIA

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RINGKASAN

Mahasiswa Teknik Perancangan dan Konstruksi Mesin dengan fokus pada desain engineering serta kompetensi teknis di bidang mekatronika, manufaktur, dan sistem kontrol. Memiliki kemampuan dalam perancangan dan analisis komponen mesin, pembuatan serta interpretasi gambar teknik sesuai standar, dan penguasaan perangkat lunak CAD seperti AutoCAD dan SolidWorks. Memahami prinsip integrasi sistem mekanik–elektrik, dasar aktuator dan sensor, serta konsep sistem kontrol dasar. Memiliki pengetahuan proses manufaktur, pemilihan material, dan pertimbangan manufacturability dalam desain. Terbiasa bekerja secara analitis, teliti, dan berorientasi pada solusi teknis yang efisien dan aplikatif.

PENGALAMAN PROYEK

GorolongBox (Mechanical Team)
- Mechanical Assembly & Maintenance
- Drivetrain & Chassis Support
- Competition Technical Support

CVT Pulley Failure Analysis for Mio 2008
- Dimensional Measurement & Inspection
- 3D Modeling & Reconstruction
- CAE & Stress Analysis
- Failure Analysis

Reverse Engineering Powder Metallurgy Mixer
- Reverse Engineering & 3D Modeling
- Structural Analysis (CAE)
- CAE & Stress Analysis
- Electrical & Control System Analysis
- System Function Evaluation

PLC Material Handling System
- Electrical Wiring
- PLC Logic Development & Implementation
- System Testing & Troubleshooting

Mechanical Chopper Design and Prototype
- Manufacturing Process & Fabrication
- Component Purchasing
- Standard Part Selection
- Mechanical Assembly & Testing

Design and Prototyping Automatic Water Pump
- Planning & Concept
- Mechanical Design & 3D Modeling
- Manufacturing & Fabrication
- Mechanical Assembly
- Control System implementation

PENDIDIKAN

Politeknik Negeri Bandung — **Agustus 2023-Sekarang**
Mechanical Engineering Study Program Mechanical Design and Contruction

MAN 1 Tasikmalaya — **Juli 2020-Juni 2023**
MIPA

Pesantren KHZ Musthafa Sukamanah — **Juli 2020-Juni 2023**
Ibtidaiyyah

HARDSKILL

- Mechanical Design & Drafting
- CAE / Simulation & Analysis
- Manufacturing & Fabrication
- Mechatronics & Electrical Systems
- Structural & Load Optimization
- Prototyping & Assembly

SERTIFIKAT

- SOLIDWORKS CAD Design Associate (CSWA- MD)
- SOLIDWORKS CAD Design Profesional (CSWP-MD)
- Awareness Ahli K3 Umum
- Mini Class PLC Siemens
- Mini Class PLC Mitsubishi FX2N

SOFTWARE

- AutoCAD
- SolidWorks
- Inventor
- Mastercam X5
- Cimatron E
- MS Office

PORTOFOLIO